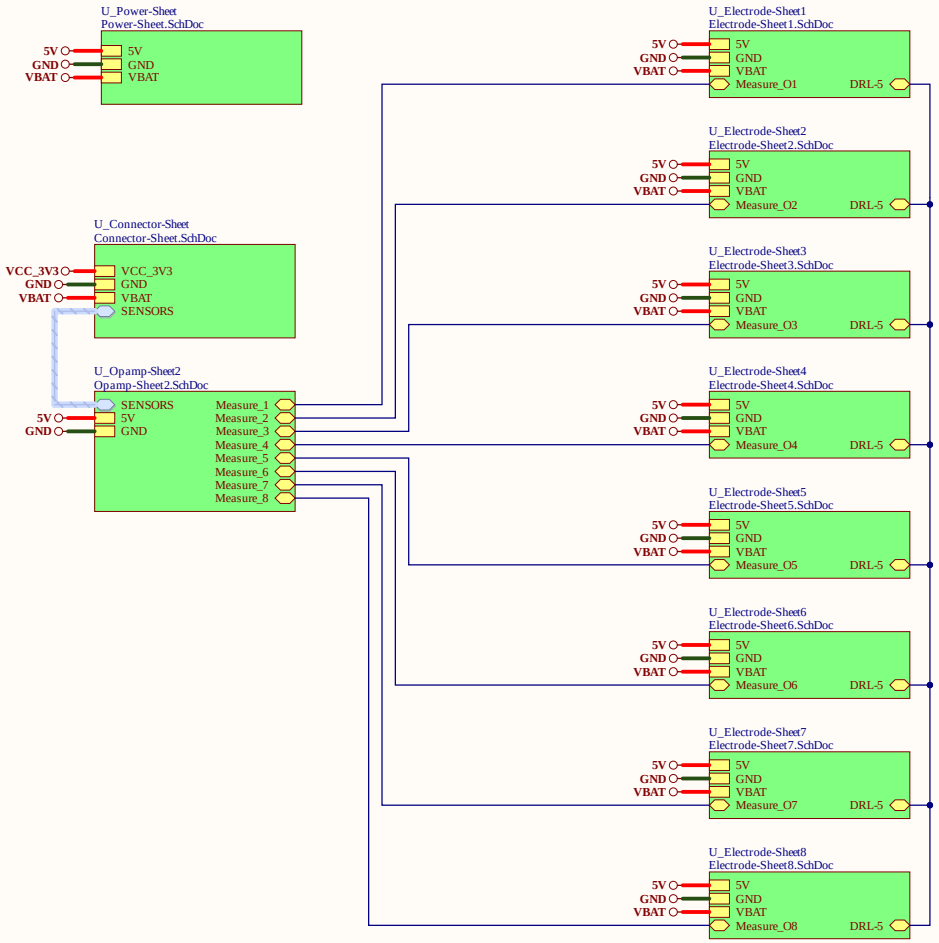
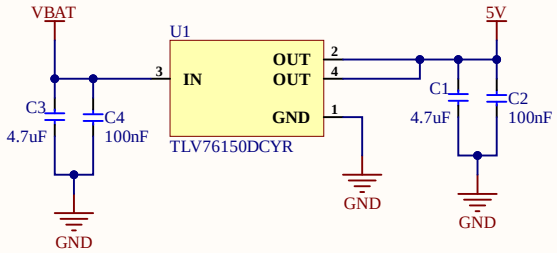
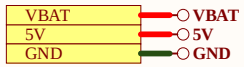


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Title: Top-Sheet.SchDoc	
File: EMG-Arm_ELC-v1.0.PrjPcb	
Date: 08/02/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 12

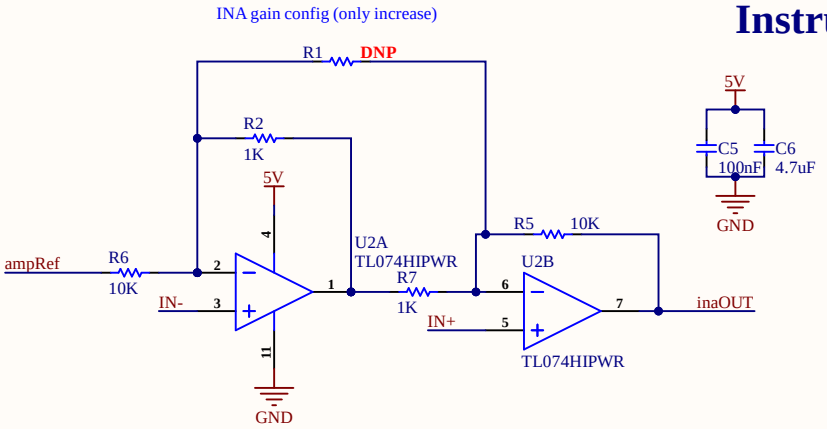


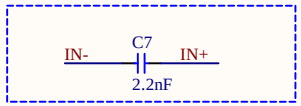


eSpark Consulting Group

Title: Power-Sheet.SchDoc	
File: EMG-Arm_ELC-v1.0.PrjPcb	
Date: 07/02/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 12

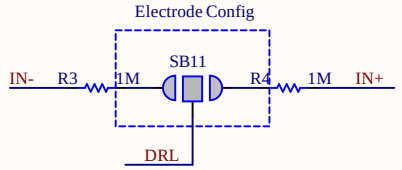
Instrumentation Amp (INA)





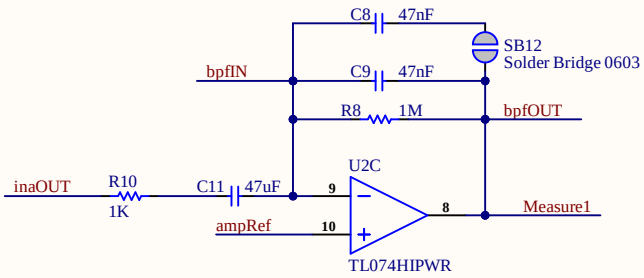
Removes high frequency noise.

Electrode Refrence

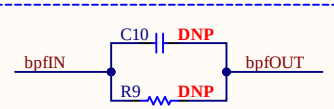


Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)



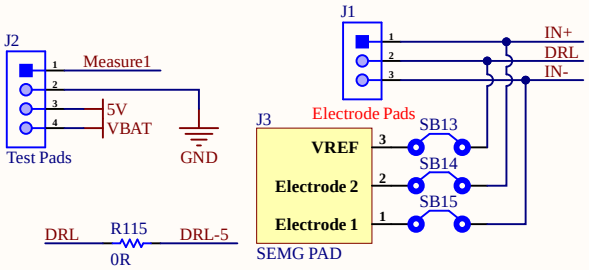
Band Pass Filter Configuration (BPFC)



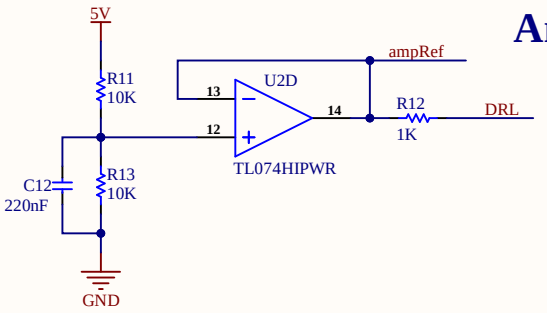
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

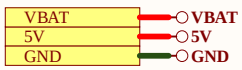
Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)



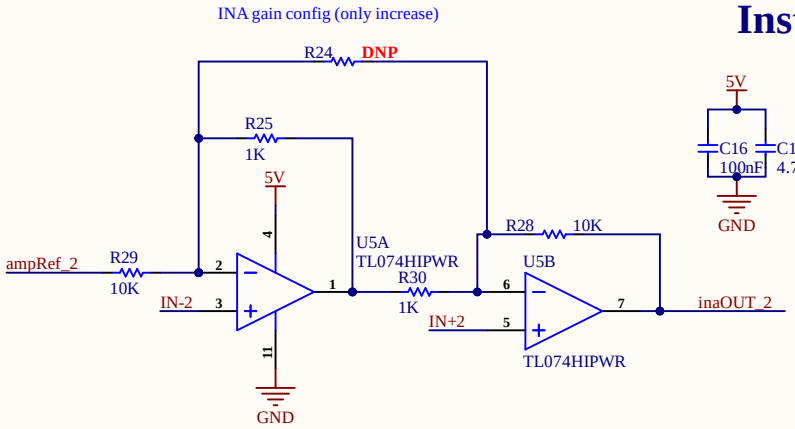


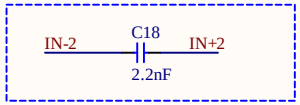


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Title: Electrode-Sheet1.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page	Num: 3
Drawn By: Huzaifa Ali	Page	Total: 12

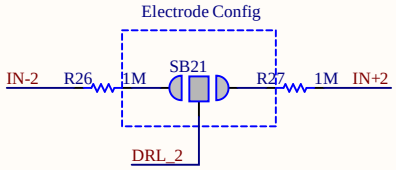
Instrumentation Amp (INA)





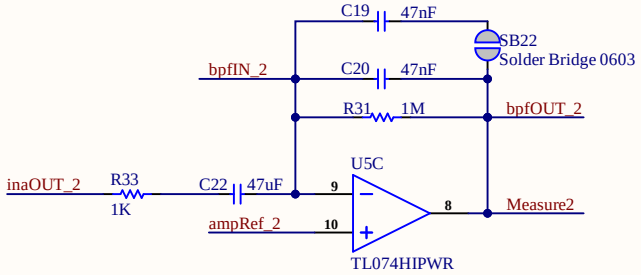
Removes high frequency noise.

Electrode Refrence

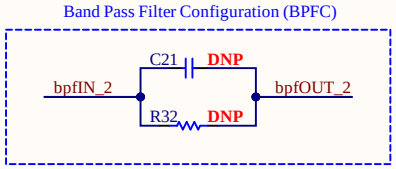


Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

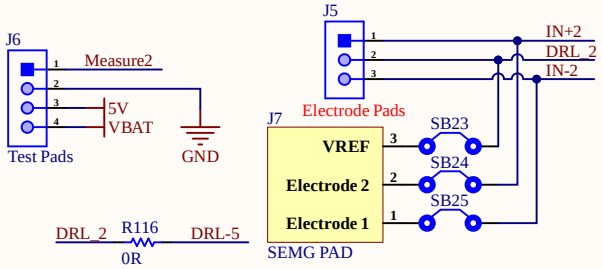


Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

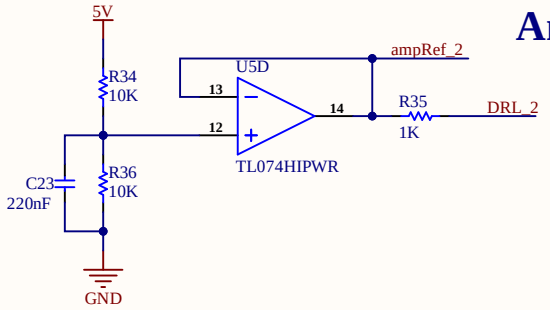


C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

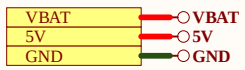
Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)



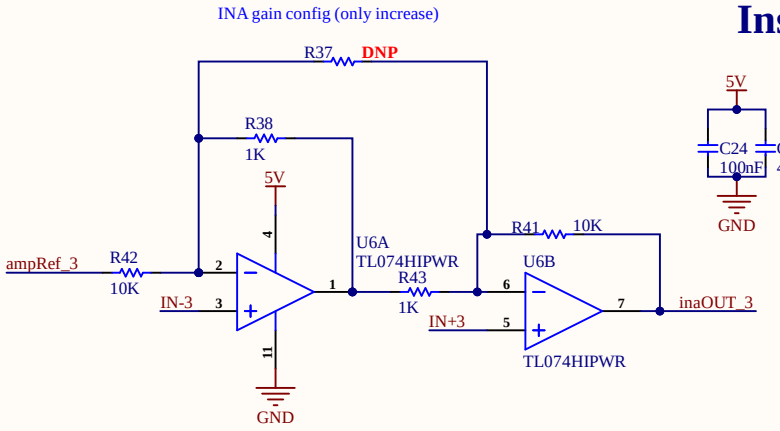


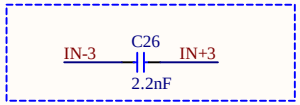


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Title: Electrode-Sheet2.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page Num:	4
Drawn By: Huzaifa Ali	Page Total:	12

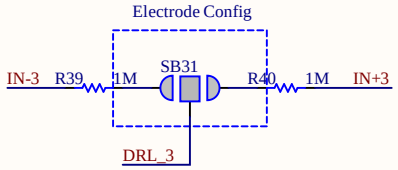
Instrumentation Amp (INA)





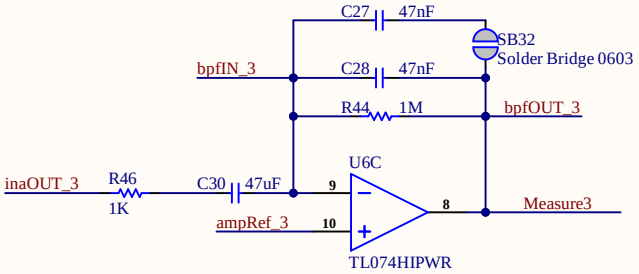
Removes high frequency noise.

Electrode Refrence

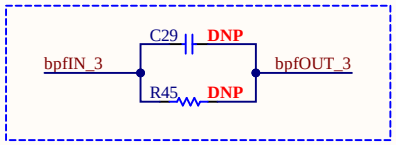


Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

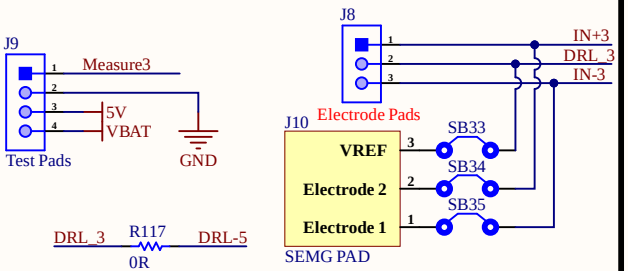


Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

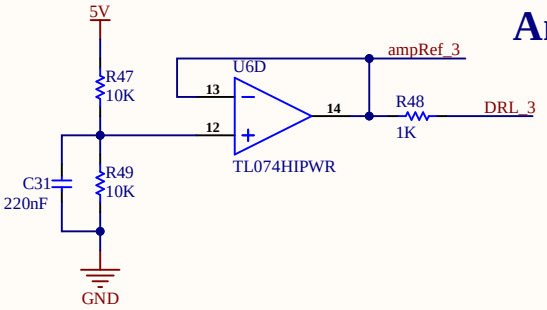


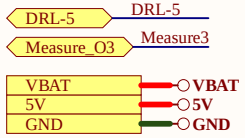
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)

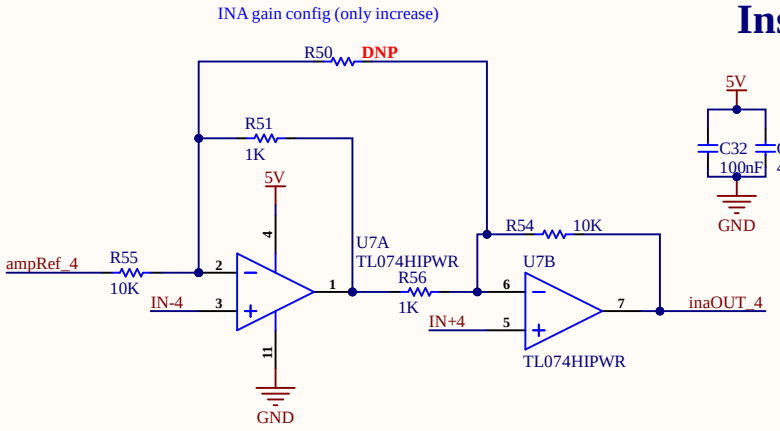


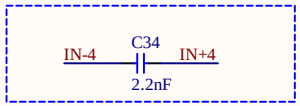


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Title: Electrode-Sheet3.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 12

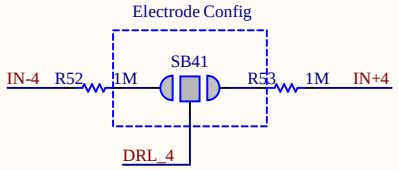
Instrumentation Amp (INA)





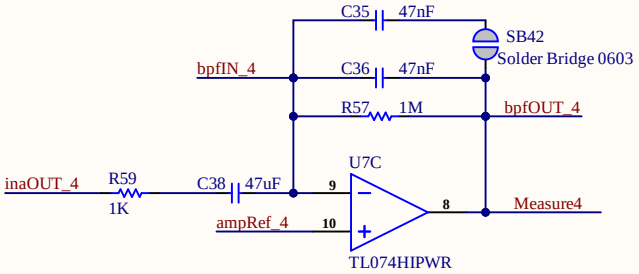
Removes high frequency noise.

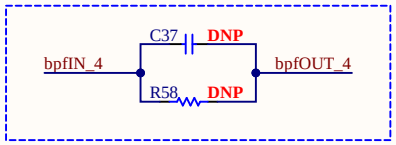
Electrode Refrence



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

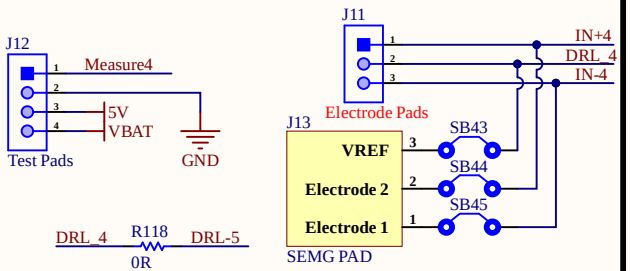




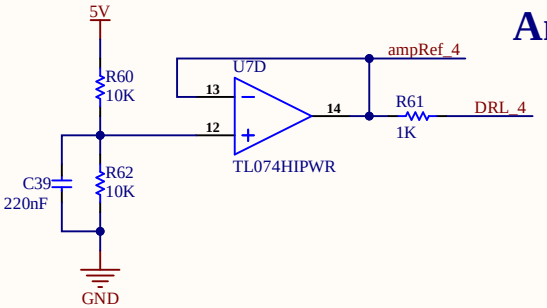
Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

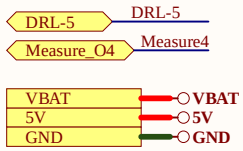
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)

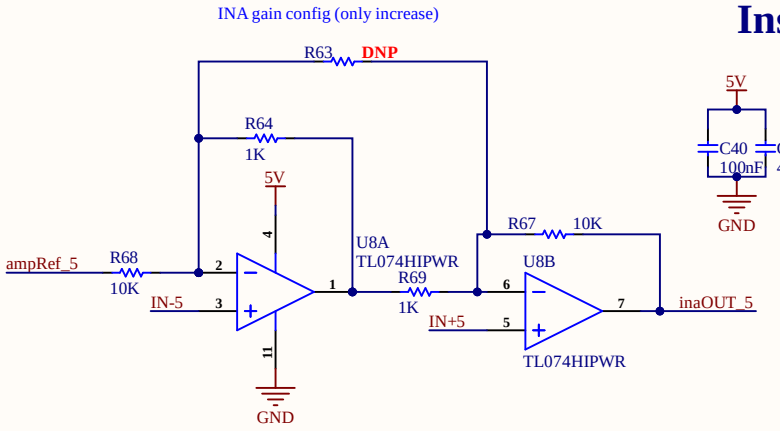


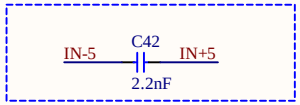


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Title: Electrode-Sheet4.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page	Num: 6
Drawn By: Huzaifa Ali	Page	Total: 12

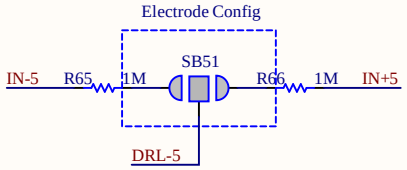
Instrumentation Amp (INA)





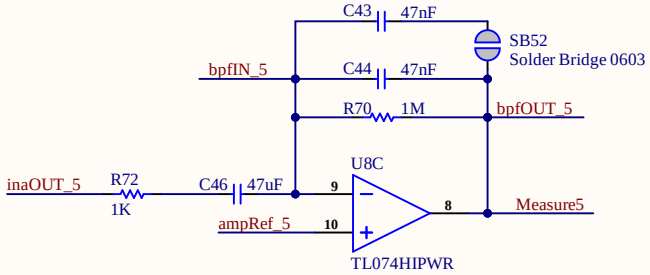
Removes high frequency noise.

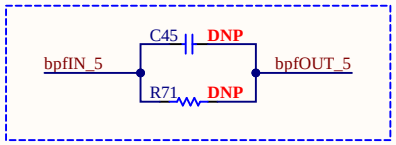
Electrode Refrence



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

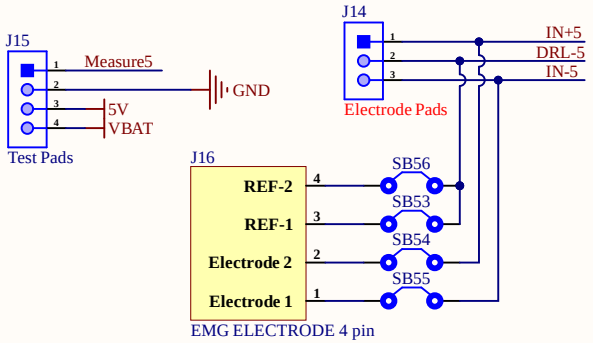




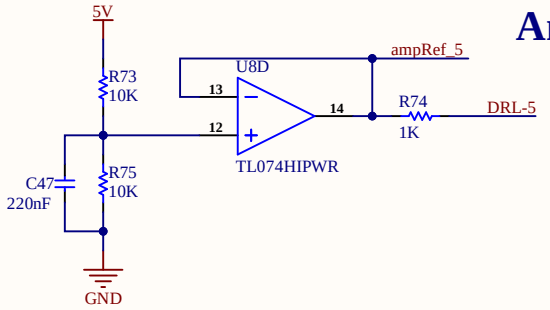
Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

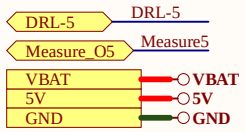
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)

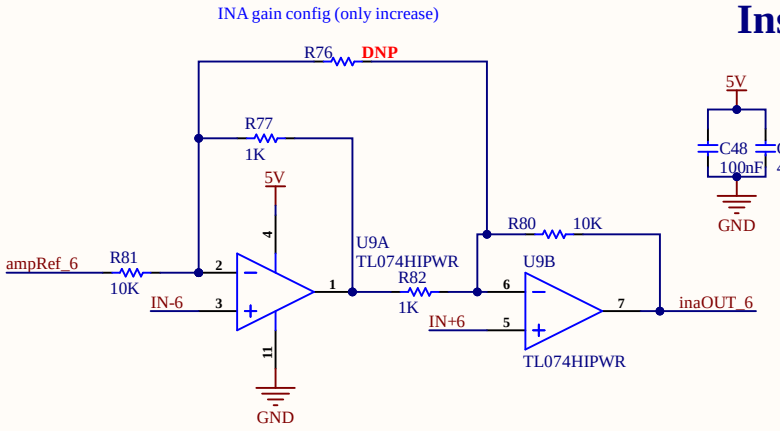


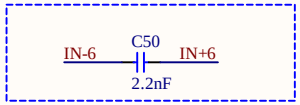


eSpark Consulting Group

Title: Electrode-Sheet5.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page	Num: 7
Drawn By: Huzaifa Ali	Page	Total: 12

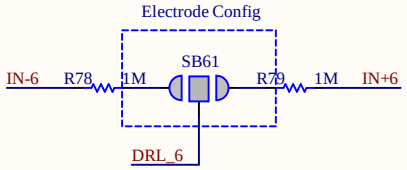
Instrumentation Amp (INA)





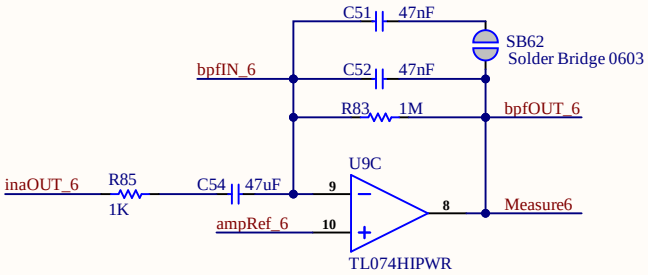
Removes high frequency noise.

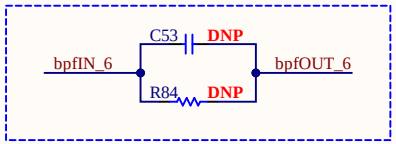
Electrode Refrence



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

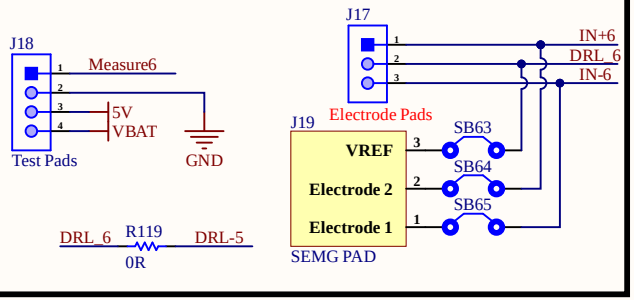




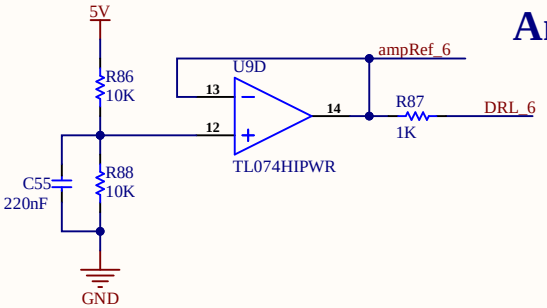
Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

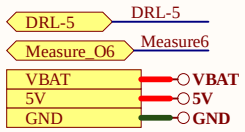
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Header Pins / Connectors



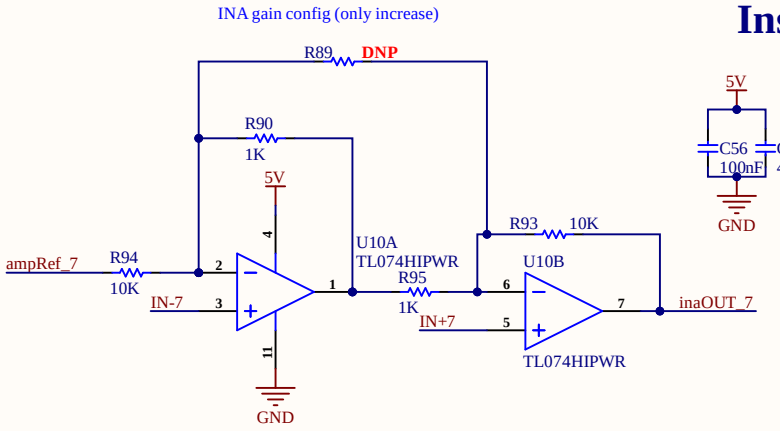
Amp Ref + Driven Right Leg (DRL)

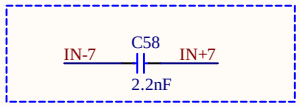




Title: Electrode-Sheet6.SchDoc	
File: EMG-Arm_ELC-v1.0.PrjPcb	
Date: 19/02/2025	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 12

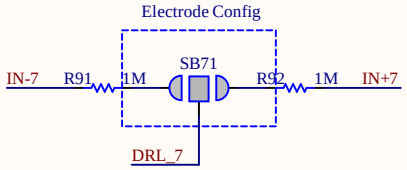
Instrumentation Amp (INA)





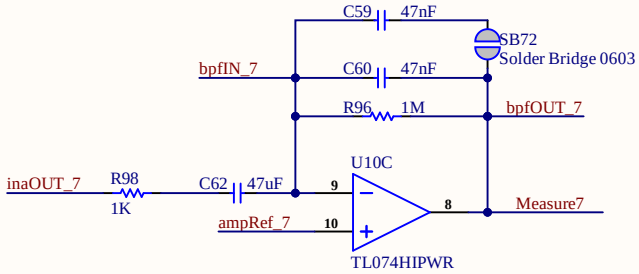
Removes high frequency noise.

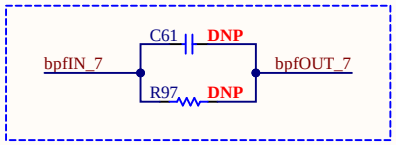
Electrode Refrence



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

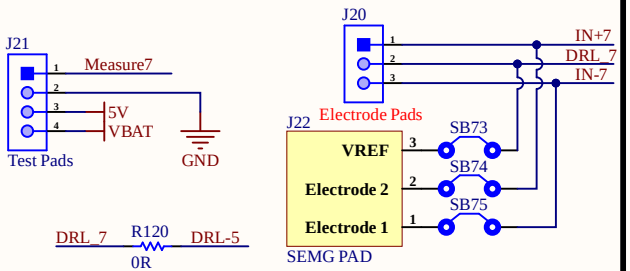




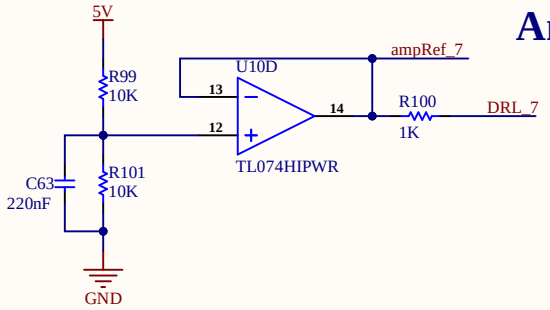
Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

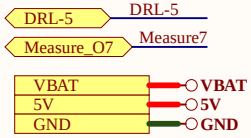
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)

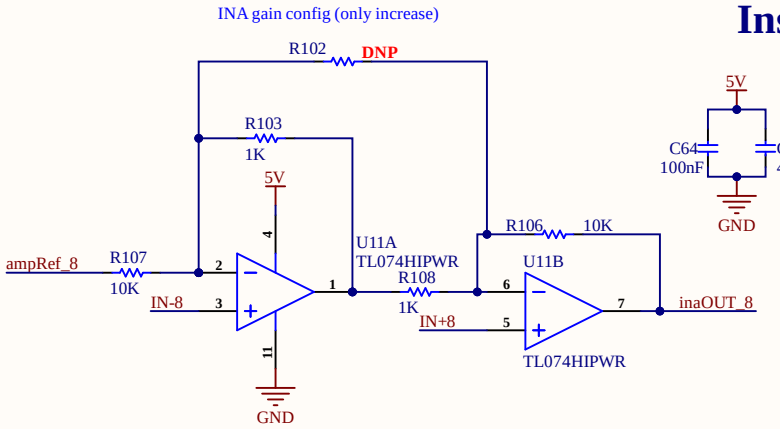


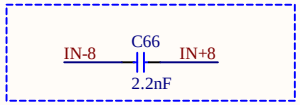


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Title: Electrode-Sheet7.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page	Num: 9
Drawn By: Huzaifa Ali	Page	Total: 12

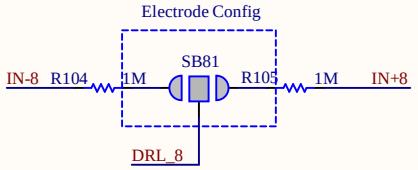
Instrumentation Amp (INA)





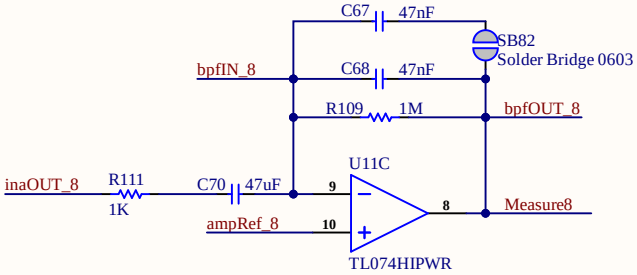
Removes high frequency noise.

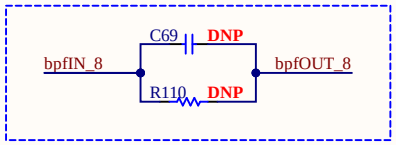
Electrode Refrence



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

1000x Gain (Bandpass)

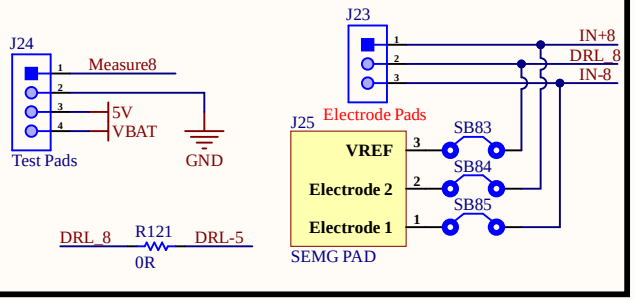




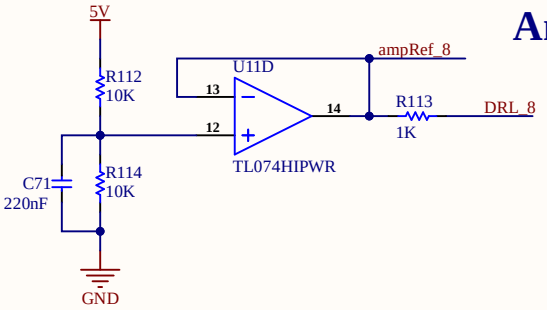
Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

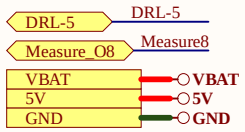
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

Header Pins / Connectors



Amp Ref + Driven Right Leg (DRL)

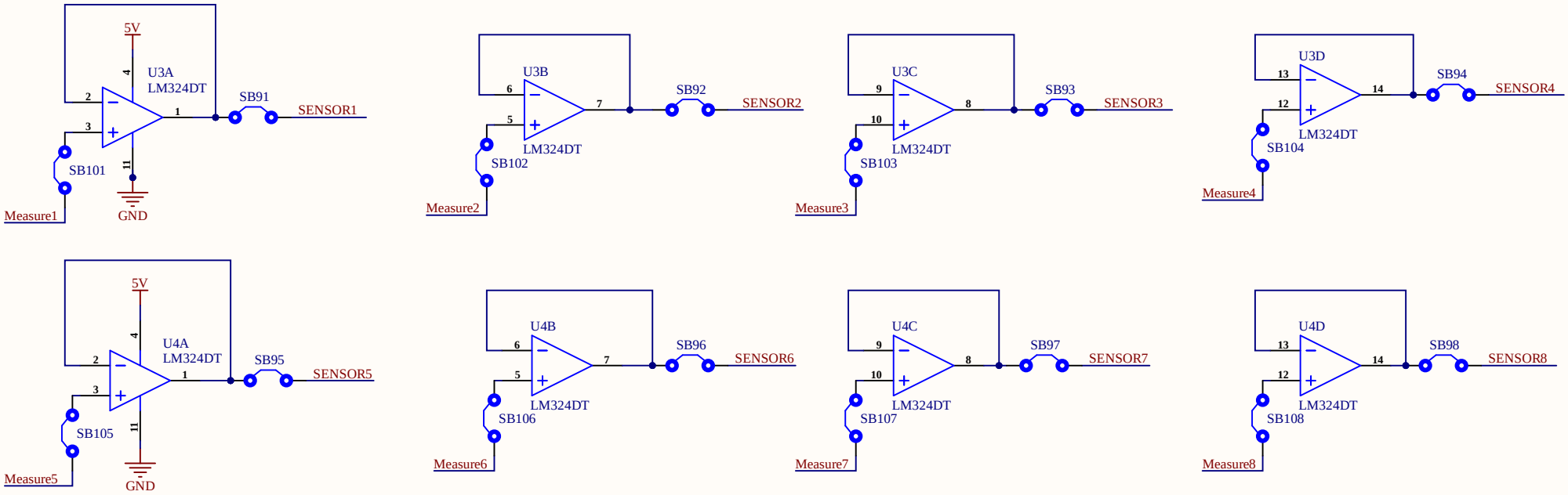




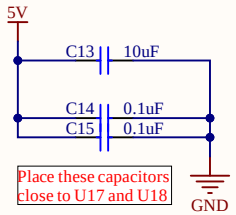
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Title: Electrode-Sheet8.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page	Num: 10
Drawn By: Huzaifa Ali	Page	Total: 12

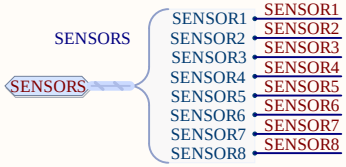
UNITY GAIN AMPLIFIER FOR IMPEDANCE MATCHING



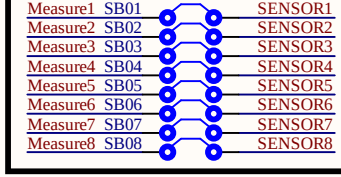
DECOUPLING CAPS

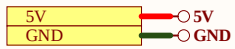


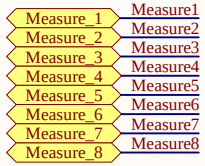
SESNOR OUT SIGNALS



SOLDER BRIDGE



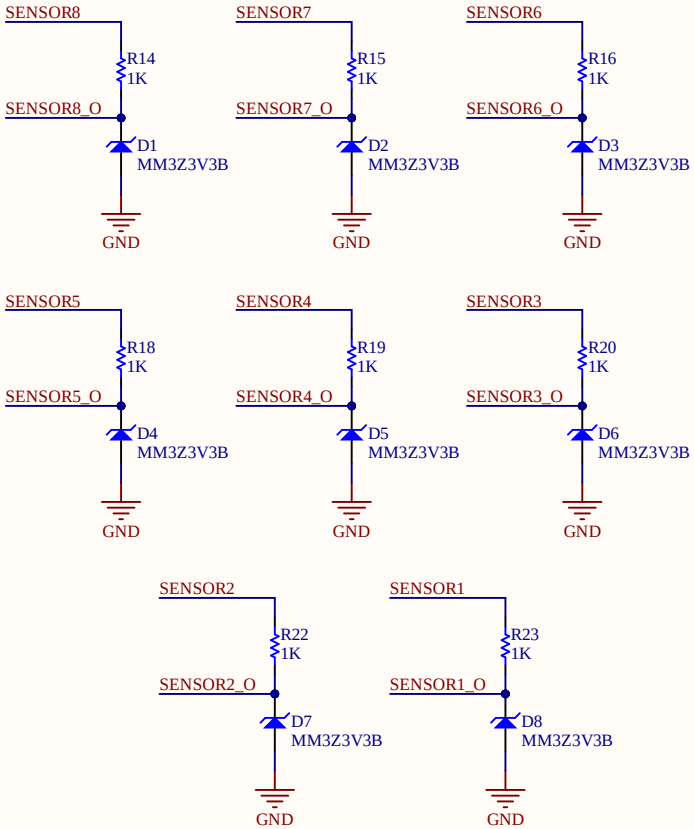




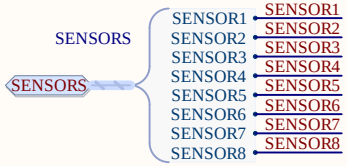
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Title: Opamp-Sheet2.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 19/02/2025	Page Num:	11
Drawn By: Huzaifa Ali	Page Total:	12

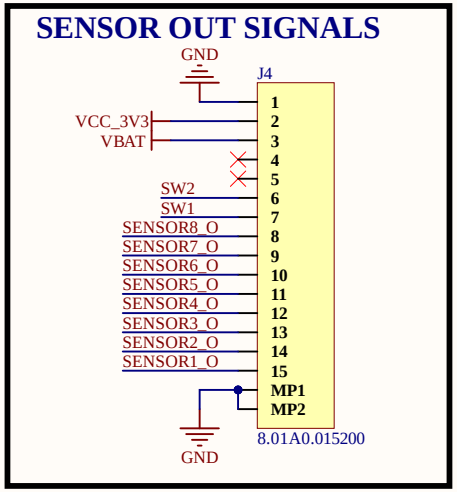
OVER VOLTAGE PROTECTION



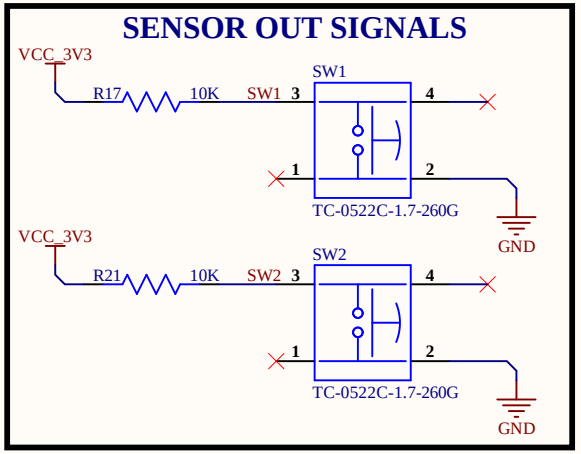
SESNOR OUT SIGNALS

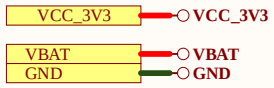


SENSOR OUT SIGNALS



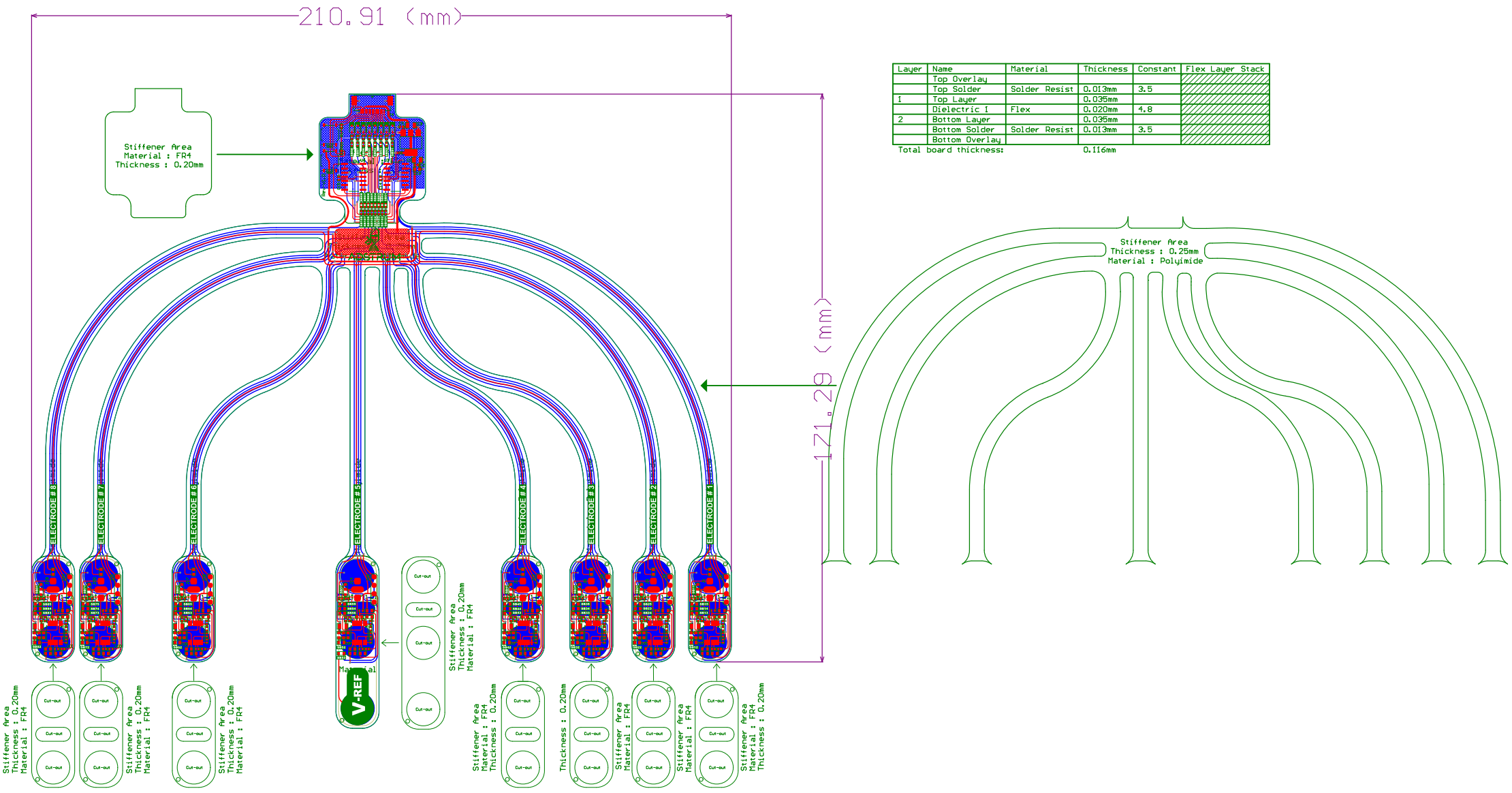
SENSOR OUT SIGNALS





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Title: Connector-Sheet.SchDoc		
File: EMG-Arm_ELC-v1.0.PrjPcb		
Date: 17/02/2025	Page	Num: 12
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Layer	Name	Material	Thickness	Constant	Flex Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.013mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 1	Flex	0.020mm	4.8	
2	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.013mm	3.5	
	Bottom Overlay				
Total board thickness:			0.116mm		